

LOOPS -- in programming world some time we keep on repeating , may be you want to repeat 5 statement so one way is copy & paste multiple times or other way is # if you want to print the datascience 1000 times then what you will you cant copy for 1000 times , if you want to print 1000 times then you cant do manually . that is the reason why we need to apply loop -> # 2 type of loops -- While loop & For loop

```
In [1]: print('data science')
        print('data science')
        print('data science')
        print('data science')
        print('data science')
```

```
data science
data science
data science
data science
data science
```

```
In [2]: i = 1          # initializing

        while i<=5:    # condition
            print('data science')
            i = i + 1  # increment
```

```
data science
data science
data science
data science
data science
```

```
In [3]: i = 1          # initializing
        while i<=5:    # condition
            print('data science : ', i)
            i = i + 1
```

```
data science : 1
data science : 2
data science : 3
data science : 4
data science : 5
```

```
In [4]: i = 5          # initializing

        while i>=1:    # condition
            print('data science')
            i = i - 1  # decrement
```

```
data science
data science
data science
data science
data science
```

```
In [5]: i = 1

        while i<=5:
            print('Data Science')    # when we mention end then new line will not create
```

```

j = 1
while j<=4:
    print('Technology')
    j = j + 1

i = i + 1
print()

```

the output which we got is very lengthy but how to make them one line Lets re

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

```

In [6]: i = 1
        while i<=5:
            print(' datascience', end = "") # when we mention end then new line will not cr
            j = 1
            while j<=4:
                print(' technology', end="")
                j = j + 1

            i = i + 1
            print()

```

datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology

In [7]: *# Lets use while loop usig some numbers*

```
i = 1

while i <= 2 :
    j = 0
    while j <= 2 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

0 1 2
0 2 4

In [112... *# Lets use while loop usig some numbers*

```
i = 1
while i <= 4 :
    j = 0
    while j <= 3 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12

FOR LOOP - # normally while loop it work with iteration or certaion some condition but for loop it will work with sequence (list, string,int)

In [10]: name = 'nit'

```
for i in name:
    print(i)
```

n
i
t

In [11]: name1 = [1,3.2,'Hello'] *# I want print the value individually*

```
for i in name1:
    print(i)
```

1
3.2
Hello

In [12]:

```
for i in [2, 3, 7.8, 'hi']:
    print(i)
```

2
3
7.8
hi

```
In [13]: range(5)
```

```
Out[13]: range(0, 5)
```

```
In [14]: for i in range(5):  
         print(i)
```

0
1
2
3
4

```
In [15]: for i in range(2,5):  
         print(i)
```

2
3
4

```
In [16]: for i in range(1,10,3):  
         print(i)
```

1
4
7

```
In [17]: # print the value which is divisible by 5  
  
         for i in range(1,21):  
             print(i)
```

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

In [18]: *# print the value which is divisible by 5*

```
for i in range(1,51):
```

```
    if i%5==0:  
        print(i)
```

5
10
15
20
25
30
35
40
45
50

In [19]: *# print the value which is divisible by 5 i dont want that value*

```
for i in range(1,51):
```

```
    if i%5!=0 :  
        print(i)
```

1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
21
22
23
24
26
27
28
29
31
32
33
34
36
37
38
39
41
42
43
44
46
47
48
49

In [20]: *# write the code user ask chocklet from vendor machne write the basic code*

```
x = int(input('How many choclets you want:'))  
  
i = 1  
while i<=x:  
    print('choclet')  
    i += 1
```

```
choclet
choclet
choclet
choclet
choclet
choclet
choclet
choclet
choclet
choclet
```

- If the user says i need 10 choclet but vending machine dont have 10 chocolate & machine has only 5 chocolate so what you do on those scenario - We have 3 choice now (eiter stop the transaction by you or you can give only 5 chocolate) & may be vendor machine display the result as we are out of the stock - Now lets try in the code

```
In [ ]: ava = 5 # the machine has only 5 choclet

x = int(input('How many choclets you want:?'))

i = 1
while i<=x:
    print('choclet')
    i += 1
# if you check the user wants 10 choclets but availabe choclet is 5 but we got out
# in this code we just declare but we dint apply any condition to it
```

```
In [22]: available_choclet = 5 # the machine has only 10 candis

x = int(input('How many choclets user want:?'))

i = 1
while i<=x:

    if i>available_choclet: # we stop the execution but which code execution not en
        break # break is statement | means jump out of the loop
    print('choclet')
    i += 1

print('bye for now')
```

```
choclet
choclet
choclet
choclet
choclet
bye for now
```

```
In [23]: available_choclet = 5 # the machine has only 10 candis

x = int(input('How many choclets you want:?'))

i = 1
while i<=x:

    if i>available_choclet: # we stop the execution but which code execution not en
        print('out of stock')
        break # break is statement | means jump out of the loop
```

```
print('choclet')
i += 1

print('bye for now')
```

choclet
choclet
choclet
choclet
choclet
out of stock
bye for now

```
In [24]: for i in range(1,11):
        print(i)
```

1
2
3
4
5
6
7
8
9
10

- i dont want 11 number i want only 5 number for the range of 1 to 10

```
In [25]: for i in range(1,11):
        if i == 6:
            break
        print(i)
```

1
2
3
4
5

- in continue loop wont be terminate & exclue the assign number it give you entire output

```
In [26]: for i in range(1,11):
        if i == 3:
            continue
        print(i)
```

1
2
4
5
6
7
8
9
10

```
In [27]: for i in range(1,11):
        if i == 6:
```



```
        continue
    print('hello :',i)
```

```
hello : 1
hello : 2
hello : 3
hello : 4
hello : 5
hello : 7
hello : 8
hello : 9
hello : 10
```

In [111... *#PASS Statement - pass the code & it wont go (code give you the error)*

In [28]: `for i in range(1,11):`

```
Cell In[28], line 1
    for i in range(1,11):
        ^
SyntaxError: incomplete input
```

In [29]: `for i in range(1,11):`
`pass`

you need to print the number from 1 to 50 but dont print the number which is divisible by 3 or 5

In [31]: `for i in range(1,51):`

 `if i%3 == 0:`

 `print(i)`
`print('end')`

```
3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end
```

In [32]: `for i in range(1,51):`

 `if i%3 == 0:`
 `continue`

```
print(i)
print('end')
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end
```

In [33]: `for i in range(1,51):`

```
    if i%3 == 0 or i%5 == 0:
        continue
    print(i)
```

it will skip all the value which is divisible by 3 or 5

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49

```
In [34]: for i in range(1,50):  
        if i%3 == 0 or i%5 == 0:  
            continue  
        print(i)  
print('end')  
# when you apply and you wont get the value which is divisible by both 3 & 5 (15)
```

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end

In [35]: *# i dont want to print the values which are even numbers that means print only odd*

```
for i in range(1,51):  
    if (i%2 == 0):  
        #print('even')  
        continue  
    else:  
        print(i)  
print('bye')
```

1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
bye

PRINTING PATTERN IN PYTHON

```
In [36]: print('# # # #')  
print('# # # #')  
print('# # # #')  
print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [37]: for i in range(1,5):  
        i=i+1  
        print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [40]: for i in range(1,5):  
        if i<=5:  
            print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [41]: for j in range (4):  
         print('#')
```

```
#  
#  
#  
#
```

```
In [47]: for j in range(4):  
         print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [50]: for j in range(4):  
         print('#', end=" ")  
  
         for j in range (4):  
             print('#', end=" ")
```

```
# # # # # # # #
```

```
In [51]: for j in range(4):  
         print('#', end=" ")  
  
         print()  
  
         for j in range(4):  
             print('#', end=" ")
```

```
# # # #  
# # # #
```

```
In [52]: for j in range(4):  
         print('#', end=" ")  
  
         print()  
  
         for j in range(4):  
             print('#', end=" ")  
  
         print()  
  
         for j in range(4):  
             print('#', end=" ")  
  
         print()
```

```
for j in range(4):
    print('#', end=" ")
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [53]: for i in range(4):                # use debug mode in pycharm
        for j in range(4):
            print('#', end=" ")
        print()
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [60]: for i in range(4):
        for j in range(i+1):
            print('#', end=" ")
        print()
```

```
#
# #
# # #
# # # #
```

```
In [63]: for i in range(1,5):
        print(" # "*i)
```

```
#
# #
# # #
# # # #
```

```
In [64]: for i in range(1,5):
        for j in range(4):
            if i>j:
                print("#",end=" ")
        print()
```

```
#
# #
# # #
# # # #
```

```
In [66]: list(range(5))
```

```
Out[66]: [0, 1, 2, 3, 4]
```

```
In [69]: for i in range(4):
        for j in range(i+1):
            print('#', end=" ")
        print()
```

```
#
# #
# # #
# # # #
```

```
In [70]: for i in range(1,5):
         for j in range(4):
             if i>j:
                 print("#",end=" ")
         print()
```

```
#
# #
# # #
# # # #
```

```
In [71]: for i in range(4):
         for j in range(i+1):
             print('#', end=" ")
         print()
```

```
#
# #
# # #
# # # #
```

```
In [72]: for i in range(4):
         for j in range(4-i):
             print('#', end=" ")
         print()
```

```
# # # #
# # #
# #
#
```

```
In [74]: for i in range(1,5):
         print("# "*(5-i))
```

```
# # # #
# # #
# #
#
```

For Else

```
In [79]: nums = [12,15,18,21,26,30,40]

         for num in nums:
             if num %5 == 0:
                 print(num)
```

```
15
30
40
```



```
In [80]: nums = [12,45,58,41,21,40]
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)
```

45

40

```
In [84]: nums = [12,14,18,21,25,20]
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)
```

25

20

```
In [85]: nums = [12,14,18,21,20,25]
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break           # it will print only 1 number then it break
```

20

```
In [86]: nums = [12,14,18,21,20,25]
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break
```

20

```
In [ ]: nums = [10,14,18,21,5,10]
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break
```

```
In [87]: nums = [7,14,18,21,23,27] # hear there is no number which is divisible by 5 we got
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break
```

```
In [93]: nums = [1,14,18,21,23,27,29] # hear there is no number which is divisible by 5 we g
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)
```

```
        break
    else:
        print('Number Not Found') # every iteration it cheking condition
```

Number Not Found
Number Not Found
Number Not Found
Number Not Found
Number Not Found
Number Not Found
Number Not Found

In [91]: `nums = [7,14,18,21,23,27]` # hear there is no number which is divisible by 5 we got

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
else:
    print('Number Not Found') # hear else we dont write in if block but we can
```

Number Not Found

In [94]: `nums = [10,14,18,21,20,27]` #hear there is no number which is divisible by 5 we got

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
else:
    print('Not Found')
```